06 - Model Selection

Customer Churn Prediction | Repository-based interview study guide

Source: https://github.com/YuvarajGarla/Customer-churn-prediction

1. Algorithms used

The repository experiments with DecisionTreeClassifier and RandomForestClassifier. It also tests a Random Forest after PCA. The notebook does not show a Logistic Regression baseline or a formal model-search procedure.

2. Decision Tree

A Decision Tree recursively partitions the feature space using feature-threshold or category-based rules. Each internal node tests a feature condition; a terminal leaf provides a class prediction or class-probability estimate.

Gini impurity

$$\text{Gini} = 1 - \Sigma(p_k^2)$$

For binary classification, Gini = 1 - p_0^2 - p_1^2. A pure node has impurity 0. A 50:50 binary node has impurity 0.5. At each split, the algorithm seeks child nodes with low weighted impurity.

$$\text{Weighted split impurity} = (N_L/N)\ \text{Gini_L} + (N_R/N)\ \text{Gini_R}$$

Repository hyperparameters

```
DecisionTreeClassifier(
   criterion="gini", random_state=100,
   max_depth=6, min_samples_leaf=8
)
```

Parameter	Value	Meaning
criterion	gini	Impurity measure used to choose splits
random_state	100	Controls randomized behaviour
max_depth	6	Limits maximum tree depth
min_samples_leaf	8	Minimum samples required in a leaf

Strengths and limitations

- Interpretable rule structure and visualizable decisions.
- Captures nonlinear relationships and feature interactions.
- Does not generally require scaling.
- A single tree can have high variance and overfit; small data changes can produce a different tree.

3. Random Forest

Random Forest is an ensemble of decision trees. Each tree is trained using a bootstrap sample of the training observations, and at each split a random subset of features is considered. For classification, the trees' outputs are aggregated, commonly by majority vote.

Why the ensemble helps

Individual trees can be unstable. Combining sufficiently diverse trees can reduce variance and improve stability. The benefit depends on tree diversity and model settings; Random Forest is not guaranteed to outperform every alternative.

Repository hyperparameters

```
RandomForestClassifier(
   n_estimators=100, criterion="gini",
   random_state=100, max_depth=6,
   min_samples_leaf=8
)
```

Parameter	Value	Meaning
n_estimators	100	Number of trees
criterion	gini	Split impurity measure
random_state	100	Reproducibility
max_depth	6	Maximum depth per tree
min_samples_leaf	8	Minimum observations in a leaf

Strengths and limitations

- Often more stable than a single unconstrained tree.
- Captures nonlinear patterns and interactions.
- Can estimate feature importance.
- Less transparent than one tree; may require more memory and compute.

4. Decision Tree versus Random Forest

Aspect	Decision Tree	Random Forest
Structure	One tree	Ensemble of trees
Prediction	Single tree output	Aggregated tree outputs
Variance	Often higher	Often reduced through averaging
Interpretability	Relatively easy	More difficult
Compute	Lower	Higher

5. Why Random Forest was selected

In the repository's experiments, Random Forest with SMOTEENN had the highest reported aggregate metrics among the listed configurations. However, those resampled test results are compromised by resampling before splitting. Therefore, this is the author's reported selection, not a verified conclusion that Random Forest is superior on an untouched test set.

6. Interview-ready explanation

We experimented with Decision Tree and Random Forest classifiers. Decision Tree is an interpretable rule-based model that selects splits to reduce impurity, using Gini in this implementation. Random Forest combines multiple randomized trees through ensemble voting and can reduce variance relative to a single tree. The repository selected the Random Forest configuration based on its reported results, although the resampling leakage means a clean validation experiment is needed before claiming a reliable performance advantage.

7. Questions to prepare

Q. What is Gini impurity?

It measures node impurity based on class proportions: 1 minus the sum of squared class probabilities.

Q. What is bootstrap sampling?

Sampling observations with replacement to create a training sample for an individual tree.

Q. Why random feature subsets?

They decorrelate trees so their errors are less similar, improving the benefit of aggregation.

Q. Why can Random Forest overfit less than a single tree?

Averaging many diverse trees often reduces variance, although unconstrained forests can still overfit or fail to generalize.

Q. Does Random Forest need scaling?

Usually not; tree split ordering is invariant to monotonic feature scaling.